

Lecture 19 – Principal Component Analysis (PCA)

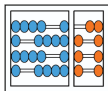
[ISLP 12.1–12.2]

MC886 – Machine Learning

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Today's summary

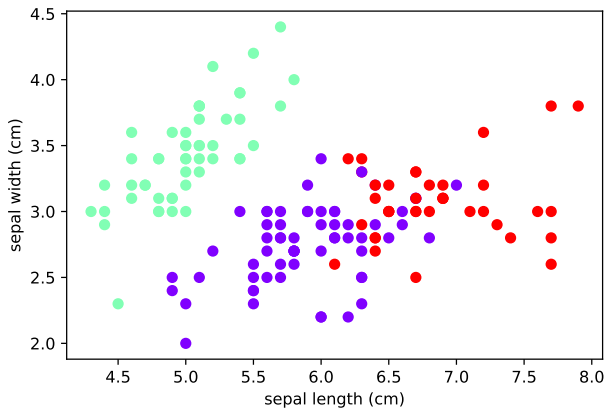
The challenge of unsupervised learning

Principal Component Analysis (PCA)

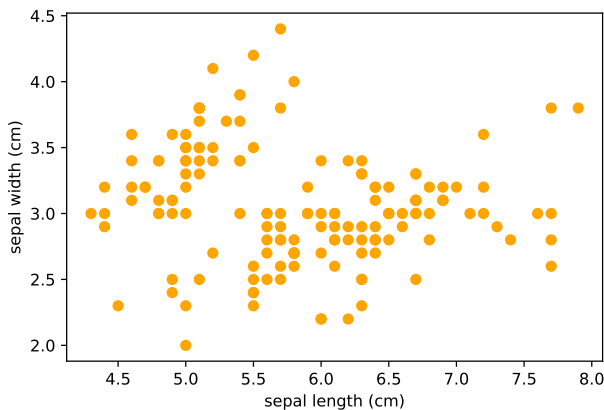
The proportion of variance explained

More on PCA

The challenge of supervised learning



The challenge of **U**nsupervised learning



$x_1, x_2, \dots, x_n$ instead of $(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)$

Reasons for Unsupervised Learning

1. Collecting and labeling patterns in big data can be costly
2. Patterns can change over the classification process
3. A more suitable data visualization (exploratory data analysis)
4. Preprocessing for supervised learning

Dimensionality reduction

- ❖ Given a dataset whose features are $X_1, \dots, X_p$, $X_j \in \mathbb{R}$, we look for a transformation Φ such as:

$$\Phi : \mathbb{R}^p \rightarrow \mathbb{R}^M,$$

where $0 < M \leq p$

- ❖ **Caveat:** Such transformation must retain as much as possible the information in the original features
- ❖ Motivation to find such Φ ?

Motivation 1: preprocessing for supervised learning

- ❖ If we have too many features, we may overfit
- ❖ Use Φ to work with fewer features
($Z_1, \dots, Z_M$ yielded by that transformation)
- ❖ Linear regression with those fewer features: “Principal Components Regression” (PCR), discussed in Section 6.3

Motivation 2: data visualization

- ❖ If $p > 3$, we can't visualize all data in a single graph
- ❖ An alternative might be to transform its p -dimensional observations into 2- or 3-dimensional vectors

Principal Component Analysis (PCA)

- ❖ PCA produces a low-dimensional representation of the dataset
- ❖ It finds a sequence of M linear combinations of the variables:

$$\begin{aligned}Z_1 &= \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p \\Z_2 &= \phi_{12}X_1 + \phi_{22}X_2 + \dots + \phi_{p2}X_p \\&\vdots \\Z_M &= \phi_{1M}X_1 + \phi_{2M}X_2 + \dots + \phi_{pM}X_p,\end{aligned}\tag{12.1}$$

such that $\sum_{j=1}^p \phi_{jm}^2 = 1$, where $1 \leq m \leq M$

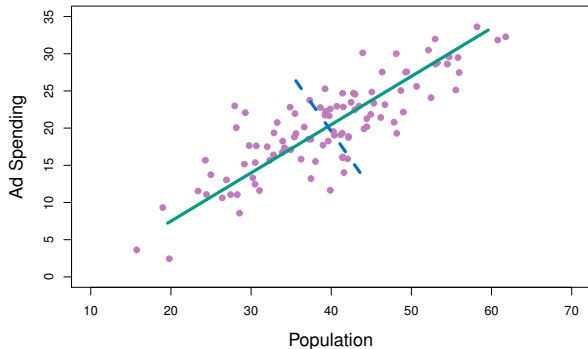
- ❖ Those linear combinations (also known as “principal components”) also have two additional properties

Properties of linear combinations in PCA

The sequence of M principal components (PCs) is constructed in a way that the m -th combination has two properties:

1. It has maximal variance
 2. It is uncorrelated to the previous $m - 1$ PCs
- ❖ For the 1st PC, the second property is not required (there is no previous principal component)

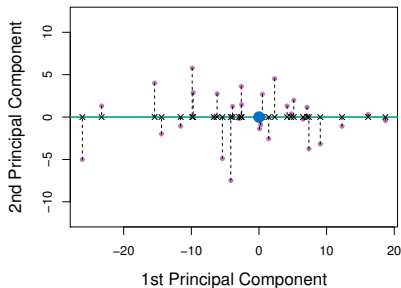
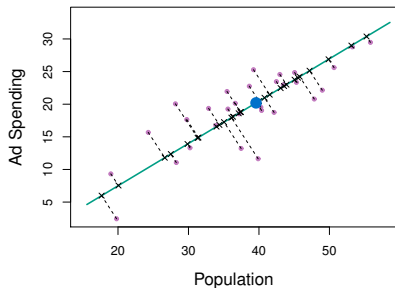
Example:
population size vs. ad spending for 100 different cities



Green solid line: 1st principal component

Blue dashed line: 2nd principal component

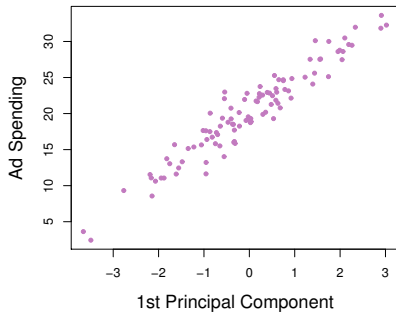
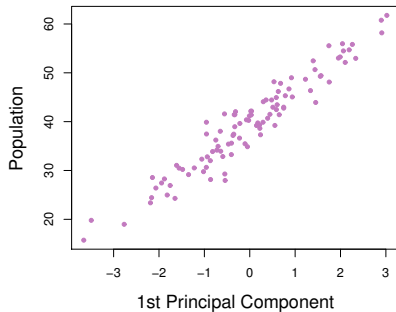
The first two principal components (PCs)



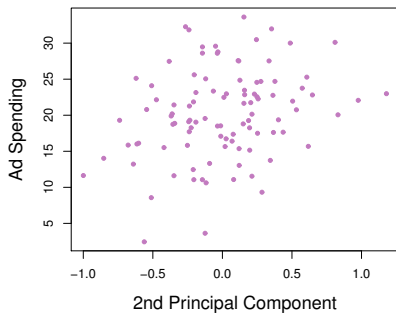
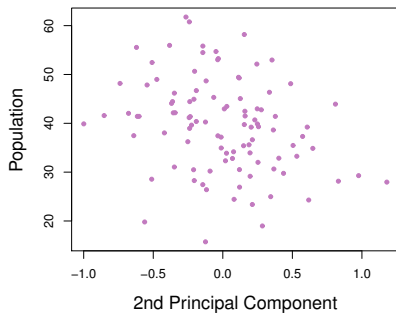
Left: 1st PC direction in green, big blue dot is the pair of averages of population and ad, dashed lines are distances of points to the 1st PC

Right: 1st PC is rotated to match the X axis

1st PC against population size and ad spending (strong relationships between 1st PC and those variables)



2nd PC against population size and ad spending (weaker relationships between 2nd PC and those variables)



Computing the first principal component

❖ Suppose we have a dataset $x_1, \dots, x_n$, $x_i = \begin{pmatrix} x_{i1} \\ \vdots \\ x_{ip} \end{pmatrix}$

❖ Moreover, assume that each feature has mean zero (if it doesn't, we can perform mean normalization)

❖ For each point x_i of the dataset, we want to compute:

$$z_{i1} = \phi_{11}x_{i1} + \phi_{21}x_{i2} + \dots + \phi_{p1}x_{ip}, \quad (12.2)$$

such that $\sum_{j=1}^p \phi_{j1}^2 = 1$

Computing the first principal component (cont.)

- ❖ For each point x_i of the dataset, we want to compute:

$$z_{i1} = \phi_{11}x_{i1} + \phi_{21}x_{i2} + \dots + \phi_{p1}x_{ip}, \quad (12.2)$$

such that $\sum_{j=1}^p \phi_{j1}^2 = 1$

- ❖ **Remark 1:** The vector $\phi_1 = (\phi_{11}, \dots, \phi_{p1})^T$ is known as the *loading vector* of the first principal component, and indicates its direction
- ❖ **Remark 2:** Z_1 also has mean zero (*why?*), hence sample variance of Z_1 is $\sum_{i=1}^n (z_{i1})^2$

Recalling PCA property 1

The sequence of M principal components (PCs) is constructed in a way that the m -th combination has two properties:

1. **It has maximal variance**
2. It is uncorrelated to the previous $m - 1$ PCs

Optimization of the first component

$$\max_{\phi_{11}, \dots, \phi_{p1}} \left\{ \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \phi_{j1} x_{ij} \right)^2 \right\}, \text{ subject to } \sum_{j=1}^p \phi_{j1}^2 = 1 \quad (12.3)$$

❖ $z_{11}, \dots, z_{n1}, z_{i1} = \sum_{j=1}^p \phi_{j1} x_{ij},$

are the values of the **1st principal component**

Recalling PCA property 2

The sequence of M principal components (PCs) is constructed in a way that the m -th combination has two properties:

1. It has maximal variance
2. **It is uncorrelated to the previous $m - 1$ PCs**

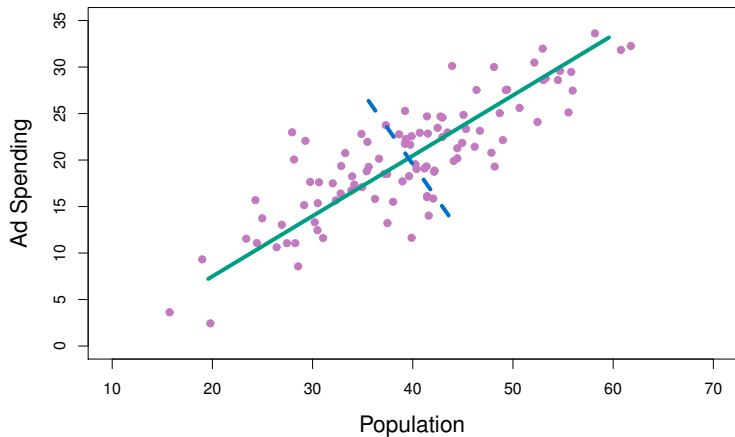
Second principal component

- ❖ The 2nd principal component is the linear combination of $X_1, \dots, X_p$ that has maximal variance among all linear combinations that are **uncorrelated** with Z_1

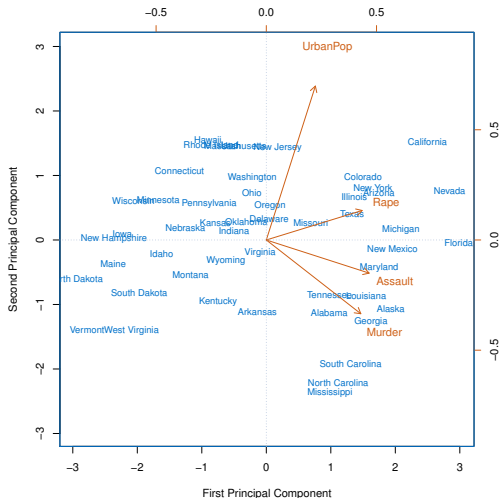
$$Z_2 = \phi_{12}X_1 + \phi_{22}X_2 + \dots + \phi_{p2}X_p$$

- ❖ Equivalently, ϕ_2 is chosen as the linear combination with maximal variance among all combinations orthogonal (perpendicular) to the direction ϕ_1
- ❖  Optimization of all $\min(n-1, p)$ principal components usually done with singular value decomposition (SVD)

Principal components are orthogonal to each other



PCA on the USArrests data ($p = 4$, $n = 50$)



Blue names: pairs (z_{i1}, z_{i2}) of transformed values (scores), $i = 1, \dots, 50$

Orange arrows: pairs (ϕ_{j1}, ϕ_{j2}) of coefficients (loadings), $j = 1, \dots, 4$

Loading vectors for the USArrests data

	PC1	PC2
Murder	0.535	-0.418
Assault	0.583	-0.187
UrbanPop	0.278	0.872
Rape	0.543	0.167

Computing $Z_1, \dots, Z_M$ with matrix algebra

❖ We compute $Z_1, \dots, Z_M$ as:

$$\begin{aligned} Z_1 &= \phi_{11}X_1 + \phi_{21}X_2 + \dots + \phi_{p1}X_p \\ &\vdots \\ Z_M &= \phi_{1M}X_1 + \phi_{2M}X_2 + \dots + \phi_{pM}X_p, \end{aligned} \tag{12.1}$$

❖ That can be expressed in matrix algebra:

$Z = \Phi X$, where:

$$Z = \begin{pmatrix} Z_1 \\ \vdots \\ Z_M \end{pmatrix}, \quad \Phi = \begin{pmatrix} \phi_{11} & \dots & \phi_{p1} \\ \vdots & \ddots & \vdots \\ \phi_{1M} & \dots & \phi_{pM} \end{pmatrix}, \quad \text{and } X = \begin{pmatrix} X_1 \\ \vdots \\ X_p \end{pmatrix}$$

Computing X from Φ and Z

- ❖ We can (approximately) recover X by computing:

$$X^T \approx Z^T \Phi$$

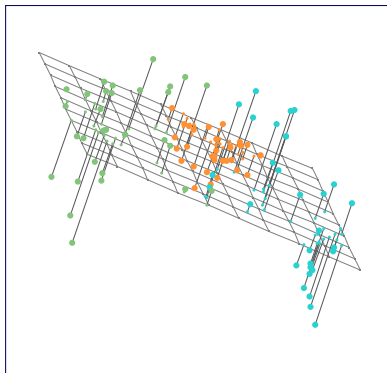
- ❖ For a single variable of X , say X_j , we have:

$$X_j \approx Z^T \phi^{(j)} = \sum_{m=1}^M Z_m \phi_{jm}, \quad \text{where } \phi^{(j)} = \begin{pmatrix} \phi_{j1} \\ \vdots \\ \phi_{jM} \end{pmatrix}$$

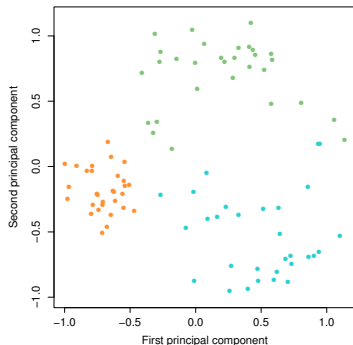
(i.e., $\phi^{(j)}$ is the j th column of Φ)

- ❖ The recovered values above are approximated unless $M = \min(n - 1, p)$ **(Why?)**

Intuitive idea through another interpretation to PCs



Left: 90 observations in 3 dimensions. The plane is positioned to minimize the distance (here, through the sum of squared distances) to each point



Right: the projection of the 90 points onto a plane (score vectors of the first two principal components)

PCs as a way to approximate x_{ij} using fewer dimensions

- ❖ We can use the first M ϕ_m and z_{im} to have the best M -dimensional approximation (in terms of Euclidean distance) to x_{ij} :

$$x_{ij} \approx \sum_{m=1}^M z_{im} \phi_{jm} \quad (12.5)$$

- ❖ When $M = \min(n - 1, p)$, the representation is exact:

$$x_{ij} = \sum_{m=1}^M z_{im} \phi_{jm}$$

How much of the information (variance) is in the first PCs?

- Given the *total variance in data*:

$$\sum_{j=1}^p \text{Var}(X_j) = \sum_{j=1}^p \frac{1}{n} \sum_{i=1}^n x_{ij}^2 \quad (12.8)$$

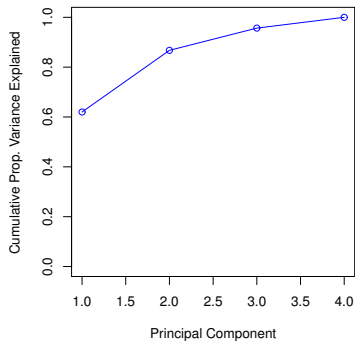
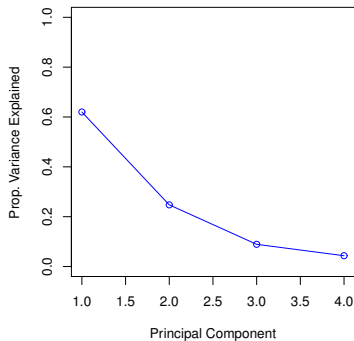
...and the variance in the *m*th principal component:

$$\frac{1}{n} \sum_{i=1}^n z_{im}^2 = \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \phi_{jm} x_{ij} \right)^2 \quad (12.9)$$

- The *proportion of variance explained* (PVE) is:

$$\frac{\sum_{i=1}^n z_{im}^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2} = \frac{\sum_{i=1}^n \left(\sum_{j=1}^p \phi_{jm} x_{ij} \right)^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2} \quad (12.10)$$

Proportion of variance explained on the USArrests data



Proportion of variance explained (PVE)

- ❖ Total variance can be decomposed into the variance of the first M PCs plus the MSE of that M -dimensional approximation:

$$\underbrace{\sum_{j=1}^p \frac{1}{n} \sum_{i=1}^n x_{ij}^2}_{\text{Var. of data}} = \underbrace{\sum_{m=1}^M \frac{1}{n} \sum_{i=1}^n z_{im}^2}_{\text{Var. of first } M \text{ PCs}} + \underbrace{\frac{1}{n} \sum_{j=1}^p \sum_{i=1}^n \left(x_{ij} - \sum_{m=1}^M z_{im} \phi_{jm} \right)^2}_{\text{MSE of } M\text{-dimensional approximation}} \quad (12.11)$$

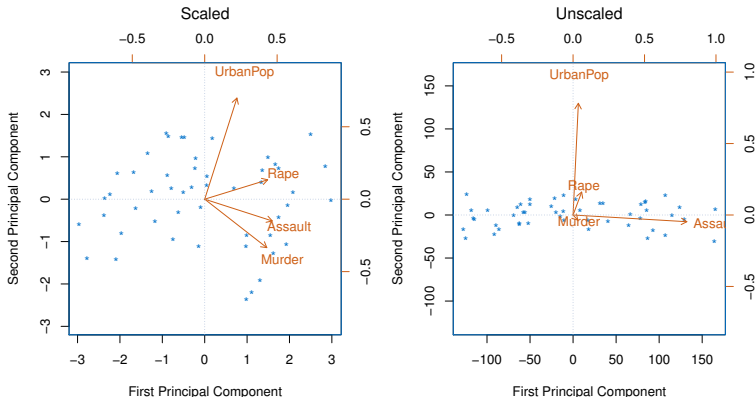
- ❖ Putting Eq. 12.11 as a function of the Var. of first PCs and replacing obtained RHS into the LHS of Eq. 12.10 yields:

$$1 - \frac{\sum_{j=1}^p \sum_{i=1}^n \left(x_{ij} - \sum_{m=1}^M z_{im} \phi_{jm} \right)^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2} = 1 - \frac{RSS}{TSS} = R^2$$

More on PCA

- ❖ The method is (highly) sensitive to variable scaling
- ❖ The method is deterministic: run PCA on two different packages will yield the same result (up to sign flips)
- ❖ How many PCs to use?
 - Visual *ad hoc* analysis
 - For PCR: cross-validation
 - For other analysis: elbow method using PVE

Scaled (left) vs. unscaled (right) variables in the USArrests data



In unscaled data, the **Assault** has by far the largest loading because it has the highest variance among variables